	LOCATION			TIME		SUB-CATCHMENT RUN OFF				INLET DESIGN										DRAIN DESIGN											HEAD LOSSES							PART FULL		DESIGN LEVELS							
				tc	I	C	A	+C x A	Q				w	d	dv		Qg	Qb		tc	I	+C x A	Qt	Qp	L	S		V	t		V2/2g	Ku	hu	Kw	hw	Sf	hf		Vp							
ARI	STRUCTURE No.	DRAIN SECTION	SUB-CATCHMENT CONTRIBUTING	SUB-CATCHMENT TIME OF CONC.	RAINFALL INTENSITY	COEFFICIENT OF RUNOFF	SUB-CATCHMENT AREA	SUM OF (C x A)	SUB-CATCHMENT DISCHARGE	FLOW IN K & C (INC. BYPASS)	ROAD GRADE AT INLET	MINOR FLOW ROAD CAPACITY	FLOW WIDTH	FLOW DEPTH	FLOW DxV	INLET TYPE	FLOW INTO INLET	BYPASS FLOW	BYPASS STRUCTURE No.	CRITICAL TIME OF CONC.	RAINFALL INTENSITY	TOTAL (C x A)	TOTAL FLOW	PIPE FLOW	REACH LENGTH	PIPE GRADE	PIPE/BOX DIMENSIONS (CLASS)	FULL FLOW VELOCITY (PIPE GRADE VELOCITY)	TIME OF FLOW IN REACH	STRUCTURE CHART No.	VELOCITY HEAD	U/S HEAD LOSS COEFFICIENT	U/S PIPE STRUCT HEAD LOSS	WSE COEFFICIENT	CHANGE IN WSE	PIPE FRICTION SLOPE	PIPE FRICTION HEAD LOSS (L x Sf)	DEPTH	VELOCITY	OBVERT LEVELS	DRAIN SECTION HGL	UPSTREAM HGL	WSE	SURFACE OR K & C LIP LEVEL	FREEBOARD	STRUCTURE No.
yr				min	mm/h		Ha	Ha	L/s	L/s	%	L/s	m	m	m2/s		L/s	L/s		min	mm/h	Ha	L/s	L/s	m	%	mm	m/s	min		m		m		m	%	m	m	m/s	m	m	m	m	m	m	
20 100	F11-2	F11-2 to FI2-2	0.193	5 5	239 311		0.139	0.139 0.139	92.1 119.9	92.1 119.9		92.1	7.500 7.500	0.150 0.150		900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS B HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	92.1 119.9		FI2-2	5.44 5.46	239 311	0.193 0.193		125.9 163.5	29	0.3	525 HDPE PN100 SN8	0.61 0.79	0.41 0.39		0.019 0.033	1.70 1.68	0.000 0.055	0.00 0.06	4.800 4.800	0.06 0.10	0.020 0.028	0.264 0.310	1.17 1.25	4.003 3.916	4.068 4.834	4.068	4.036	4.8	0.764	F11-2
20 100	FI2-2	FI2-2 to F13-2	0.466	5 5	239 311		0.147	0.147 0.147	97.6 127	97.6 127		97.6	7.500 7.500	0.150 0.150		900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS B HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	97.6 127		F13-2	5.86 5.85	239 311	0.466 0.466		299.1 388.8	10.3	0.3	600 HDPE PN100 SN8	1.09 1.42	0.12 0.12		0.061 0.105	1.00 1.00	0.100 0.105	0.10 0.11	4.884 4.884	0.14 0.24	0.010 0.025	0.421 0.591	1.43 1.42	3.916 3.885	4.019 4.705	4.019	3.958	4.884	0.926	FI2-2
20 100	F13-2	F13-2 to FI4-2	0.523	5 5	239 311		0.057	0.057 0.057	38 49.4	38 49.4		38	3.400 3.900	0.068 0.078		900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	38 49.4		FI4-2	5.98 5.97	239 311	0.523 0.523		334.3 434.4	35	0.3	750 HDPE PN100 SN8	0.80 1.04	0.39 0.37		0.033 0.056	2.20 2.21	0.100 0.123	0.10 0.12	4.705 4.705	0.06 0.10	0.020 0.036	0.383 0.452	1.50 1.59	3.885 3.780	3.943 4.557	3.943	3.871	4.705	0.833	F13-2
20 100	FI4-2	FI4-2 to FI5-2	0.748	5 5	239 311		0.095	0.095 0.095	62.8 81.7	62.8 81.7		62.8	4.450 5.200	0.089 0.104		900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	62.8 81.7		FI5-2	6.36 6.34	239 311	0.748 0.748		470.8 611.8	40	0.3	900 HDPE PN100 SN8	0.79 1.03	0.41 0.38		0.032 0.055	1.00 1.00	0.000 0.055	0.00 0.06	4.741 4.741	0.05 0.09	0.020 0.034	0.425 0.498	1.63 1.74	3.898 3.778	3.849 4.468	3.849	3.817	4.741	0.924	FI4-2
20 100	FI1-1	FI1-1 to FI2-1	0.14	5 5	239 311		0.05	0.050 0.050	33.3 43.4	33.3 43.4		33.3				900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	33.3 43.4	26	FI2-1	5.02 5.02	239 311	0.140 0.140		74.1 95	80.4	0.3	450 Class 2 RRJ	0.47 0.60	1.39 1.30		0.011 0.018	2.90 2.88	0.000 0.052	0.00 0.05	4.411 4.411	0.04 0.07	0.030 0.053	0.218 0.254	0.97 1.03	3.765 3.524	3.745 4.331	3.745	3.713	4.411	0.698	FI1-1
20 100	FI5-2	FI5-2 to FI6-2	1	5 5	239 311		0.09	0.090 0.090	59.8 77.8	59.8 77.8		59.8	4.350 5.050	0.087 0.101		900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	59.8 77.8		FI6-2	6.77 6.72	239 311	1.000 1.000		619.6 805.6	39.2	0.3	900 HDPE PN100 SN8	1.04 1.36	0.38 0.36		0.056 0.096	1.00 1.00	0.100 0.096	0.10 0.10	4.764 4.764	0.09 0.14	0.030 0.056	0.502 0.601	1.75 1.84	3.758 3.640	3.797 4.316	3.797	3.741	4.764	1.023	FI5-2
20 100	FI2-1	FI2-1 to FI3-1	0.298	5 5	239 311		0.068	0.068 0.068	45 58.6	45 84.5		45	3.700 5.300	0.074 0.106		900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	45 58.6		FI3-1	6.4 6.32	239 311	0.298 0.298		169.8 244.1	69.3	0.3	600 HDPE PN100 SN8	0.62 0.89	0.91 0.84		0.020 0.040	1.00 1.00	0.000 0.040	0.00 0.04	4.800 4.800	0.05 0.10	0.030 0.068	0.290 0.364	1.27 1.38	3.645 3.437	3.680 4.223	3.68	3.66	4.8	1.14	FI2-1
20 100	FI6-2	FI6-2 to FI7-2	1.182	5 5	239 311		0.103	0.103 0.103	68.1 88.6	68.1 88.6		68.1	4.700 5.450	0.094 0.109		900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	68.1 88.6		FI7-2	7.15 7.08	239 311	1.182 1.182		721.3 938	29.2	0.3	900 HDPE PN100 SN8	1.22 1.58	0.27 0.26		0.076 0.130	1.00 1.00	0.100 0.130	0.10 0.13	4.740 4.740	0.11 0.19	0.030 0.056	0.555 0.682	1.80 1.88	3.620 3.532	3.707 4.130	3.707	3.631	4.74	1.109	FI6-2
20 100	FI3-1	FI3-1 to FI4-1	0.509	5 5	239 311		0.058	0.058 0.058	38.3 49.8	38.3 49.8		38.3	3.400 3.950	0.068 0.079		900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	38.3 49.8		FI4-1	7.32 7.16	239 311	0.509 0.509		291.6 402.3	67.6	0.3	600 HDPE PN100 SN8	1.06 1.47	0.79 0.77		0.058 0.110	1.50 1.49	0.100 0.163	0.10 0.16	4.845 4.845	0.14 0.25	0.090 0.171	0.412 0.591	1.43 1.47	3.417 3.214	3.626 3.889	3.626	3.54	4.845	1.304	FI3-1
20 100	FI7-2	FI7-2 to MH8-2	1.204	5 5	239 311		0.022	0.022 0.022	14.7 19.2	14.7 19.2		14.7				300 WIDE ACO KLASSIK DRAIN 900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	14.7 19.2	6.6 9.3	MH8-2	7.42 7.34	239 311	1.204 1.204		720.9 936.8	28.9	0.3	900 HDPE PN100 SN8	1.22 1.58	0.27 0.26		0.076 0.130	1.00 1.00	0.100 0.130	0.10 0.13	4.952 4.952	0.11 0.19	0.030 0.055	0.555 0.681	1.80 1.88	3.512 3.425	3.598 3.944	3.598	3.522	4.952	1.43	FI7-2
20 100	FI4-1	FI4-1 to MH5-1	0.703	5 5	239 311		0.077	0.077 0.077	51.1 66.4	51.1 66.4		51.1	4.000 4.600	0.080 0.092		900x900 GRATED PIT WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	51.1 66.4		MH5-1	8.1 7.93	239 311	0.703 0.703		395.9 537.4	27.8	0.3	750 HDPE PN100 SN8	0.94 1.28	0.30 0.28		0.045 0.084	1.00 1.00	0.000 0.084	0.00 0.08	4.789 4.789	0.09 0.15	0.020 0.042	0.426 0.528	1.56 1.66	3.334 3.251	3.448 3.763	3.448	3.402	4.789	1.387	FI4-1
20 100	MH8-2	MH8-2 to FI6-1	1.303	5 5	239 311		0.031	0.031 0.031	20.6 26.9	27.2 36.2		20.6				900x900 GRATED PIT OVER 1500 DIA MANHOLE CLASS D COVER. REFER BCC STD DRG BSD-8021	20.6 26.9	20.8	FI6-1	7.68 7.59	239 311	1.303 1.303		765.2 993.4	33.6	0.3	900 HDPE PN100 SN8	1.29 1.67	0.31 0.33		0.085 0.146	1.40 1.44	0.100 0.210	0.10 0.21	4.715 4.715	0.13 0.21	0.040 0.072	0.578 0.869	1.83 1.68	3.405 3.304	3.489 3.662	3.489	3.367	4.715	1.348	MH8-2
20 100	MH5-1	MH5-1 to FI6-1	0.703		239 311											1050 DIA MH WITH CLASS D COVER. REFER BCC STD DRG BSD-8021			FI6-1	8.4 8.21	239 311	0.703 0.703		391 530.8	12.9	0.3	750 HDPE PN100 SN8	0.93 1.26	0.14 0.13		0.044 0.082	1.00 1.00	0.000 0.082	0.00 0.08	4.844 4.844	0.08 0.15	0.010 0.019	0.422 0.523	1.56 1.65	3.231 3.192	3.379 3.662	3.379	3.334	4.844	1.51	MH5-1
20 100	FI6-1	FI6-1 to GPT7-1	2.05	5 5	239 311		0.044	0.044 0.044	29.1 37.9	43.8 58.8		29.1				900x900 GRATED PIT OVER 1500ø MH WITH OCEAN GUARD FILTER BASKET. CLASS D HEELSAFE GRATE. REFER BCC STD DRG BSD-8091	29.1 37.9	37.4		8.54 8.34	239 311	2.050 2.050		1142.3 1508.3	3.5	0.3	1050 HDPE PN100 SN8	1.30 1.72	0.03 0.03		0.086 0.151	2.00 1.98	0.200 0.298	0.20 0.30	4.623 4.623	0.10 0.18	0.000 0.006	0.647 0.795	2.03 2.13	3.497 3.487	3.324 3.354	3.324	3.153	4.623	1.47	FI6-1
20 100	GPT7-1	GPT7-1 to HW8-1	2.05		239 311											ENVIROAUSTRALIS H90 INSTALLED IN ACCORDANCE WITH MANUFACTURER'S SPECIFICATIONS.			HW8-1	8.57 8.36	239 311	2.050 2.050		1140.8 1506.5	16	0.3	1050 HDPE PN100 SN8	1.30 1.72	0.13 0.13		0.086 0.150	1.00 1.00	0.100 0.150	0.10 0.15	4.723 4.723	0.10 0.17	0.020 0.028	0.646 0.794	2.03 2.13	3.467 3.419	3.143 3.156	3.143	3.056	4.723	1.666	GPT7-1

NOT FOR CONSTRUCTION

revision notes

DRN	AMENDMENTS	DATE	REV
JD	ISSUED FOR TENDER	23.12.25	A
JD	ISSUED FOR TENDER	28.01.26	B
LQ	ISSUED FOR TENDER	30.01.26	C

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NORTHROP
Brisbane
Level 25, 12 Creek Street, Brisbane QLD 4000
GPO BOX 423 Brisbane QLD 4001
(07) 3365 0400 brisbane@northrop.com.au
ABN 81 094 433 100

Drawn J. DENG	Checked A. DE ANDRADE
Design Coordinated R. MITCHELL	Compliance Checked S.KILLORAN
Authorised For Issue S.KILLORAN	Construction Checked N/A

SKYGATE PLAY
BRISBANE AIRPORT

STORMWATER DRAINAGE
CALCULATION TABLE

BRISBANE AIRPORT CORPORATION PTY LTD
11 THE CIRCUIT, BRISBANE AIRPORT QLD 4008
PO BOX 61, HAMILTON CENTRAL QLD 4007
+61 7 3406 3000
WWW.BNE.COM.AU

Scale	AS ABOVE
Size	A1
Revision	C

Drawing Number	Project No.	Originator	Volume	Level	Type	Discipline	Number
	25021 -	NRP -	XX -	XX -	DR-	CI	-000341

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